BME 708 — Project Plan

Real-Time Deep Learning BCI for Motor Imagery | *Giacomo Albergo* | *University of Miami*

Objective

Continue the BCI-FES pipeline built in BME 707 toward a validated, hardware-integrated prototype. The three goals for this semester are: (1) online adaptive calibration to cut setup time, (2) a working FES hardware trigger with characterised latency, and (3) a complete, mentor-approved MS Project Report.

Online Adaptive Calibration

— Implement session-to-session transfer learning (Euclidean alignment or CORAL)

— Reduce calibration time from ~30 min to under 5 min

— Online threshold drift correction via exponential moving average

FES Hardware Integration

— Connect classifier output to a hardware trigger (Arduino / DAQ board)

— Measure full end-to-end latency: EEG → classification → stimulation

— Target: under 200 ms, which is the accepted standard for real-time FES

Validation & Report

— 10-fold leave-one-session-out validation across all 11 subjects

— Benchmark comparison against Lawhern et al. 2018 and Schirrmeister et al. 2017

— Write and submit the full BME 708 MS Project Report

Success Criteria

— Calibration time < 5 min

— End-to-end latency < 200 ms

— Real-time balanced accuracy ≥ 70% mean (maintained from BME 707 results)

— Final report submitted and approved by Apr 27